



SIMATS Online Programming Lab

JAVA PROGRAMMING



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CO1-AT3-Coding Challenge

View Only

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QUESTIONS

Q1

Smart University Payroll
Management System
25 marks



Q2

Smart Vehicle Insurance Claim
System
25 marks



2/2 answered

Q2 Smart Vehicle Insurance Claim System 25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type.

Requirements Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount

Question 3 (Very Hard) Online Shopping Order Processing System (50 Marks)

Problem Statement An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. Requirements Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

$$\text{Claim} = \text{insuredAmount} \times 0.80$$

Bike

$$\text{Claim} = \text{insuredAmount} \times 0.60$$

Truck

$$\text{Claim} = \text{insuredAmount} \times 0.90$$

Test Case 1

Input

3

Car

TN10AB1234

Ramesh

800000

Bike

TN09XY5678

Suresh

100000

Truck

TN11TR9999

Logan

1500000

Search Vehicle:

TN09XY5678

Expected Output

Vehicle : TN10AB1234

Claim Amount : 640000.0

Vehicle : TN09XY5678

Claim Amount : 60000.0

Vehicle : TN11TR9999

Claim Amount : 1350000.0

Highest Claim Vehicle

TN11TR9999

1350000.0

Vehicle Found

Owner : Suresh

Claim Amount : 60000.0

Your Code

```
1 import java.util.*;
2
3 class Vehicle {
4     public static void main(String args[]){
5         private String vehicleNumber;
6         private String ownerName;
7         private double insuredAmount;
8
9         public Vehicle(String vehicleNumber, String ownerName, double insuredAmount){
10             this.vehicleNumber = vehicleNumber;
11             this.ownerName = ownerName;
12             this.insuredAmount = insuredAmount;
13         }
14
15         public String getVehicleNumber() {
16             return vehicleNumber;
17         }
18
19         public String getOwnerName() {
20             return ownerName;
21         }
22
23         public double getInsuredAmount() {
24             return insuredAmount;
25         }
26
27         public double calculateClaimAmount() {
28             return 0.0;
29         }
30     }
31
32 class Car extends Vehicle {
33     public Car(String vehicleNumber, String ownerName, double insuredAmount){
34         super(vehicleNumber, ownerName, insuredAmount);
35     }
36 }
```

```
35     }
36
37     @Override
38     public double calculateClaimAmount() {
39         return getInsuredAmount() * 0.80;
40     }
41 }
42
43 class Bike extends Vehicle {
44     public Bike(String vehicleNumber, String ownerName, insuredAmount) {
45         super(vehicleNumber, ownerName, insuredAmount);
46     }
47
48     @Override
49     public double calculateClaimAmount() {
50         return getInsuredAmount() * 0.60;
51     }
52 }
53
54 class Truck extends Vehicle {
55     public Truck(String vehicleNumber, String ownerName, insuredAmount) {
56         super(vehicleNumber, ownerName, insuredAmount);
57     }
58
59     @Override
60     public double calculateClaimAmount() {
61         return getInsuredAmount() * 0.90;
62     }
63 }
64
65 class Main {
66
67     static String fmt(double value) {
68         return String.format("%.1f", value);
69     }
70
71
72     Scanner sc = new Scanner(System.in);
73
74     int n = Integer.parseInt(sc.nextLine());
75
76     Vehicle[] vehicles = new Vehicle[n];
77
78     for (int i = 0; i < n; i++) {
79
```

```

80         while (true) {
81             String line = sc.nextLine()
82             if (!line.isEmpty()) {
83                 String type = line;
84
85                 String vehicleNumber = ;
86                 String ownerName = sc.ne
87                 double insuredAmount = 1
88
89                 if (type.equals("Car"))
90                     vehicles[i] = new Ca
91             } else if (type.equals('
92                 vehicles[i] = new B
93             } else if (type.equals('
94                 vehicles[i] = new T
95             }
96
97             break;
98         }
99     }
100 }
101
102 for (Vehicle v : vehicles) {
103     System.out.println("Vehicle : "
104     System.out.println("Claim Amount
105     System.out.println();
106 }
107
108 Vehicle highest = vehicles[0];
109
110 for (int i = 1; i < n; i++) {
111     if (vehicles[i].calculateClaimAm
112         highest = vehicles[i];
113     }
114 }
115
116 System.out.println("Highest Claim Ve
117 System.out.println(highest.getVehic
118 System.out.println(fmt(highest.calcl
119 System.out.println();
120 String line = sc.nextLine().trim();
121 if (line.equalsIgnoreCase("Search Ve
122     line = sc.nextLine().trim();
123 }
124

```

```
125         String searchNumber = line;
126
127         Vehicle found = null;
128
129         for (Vehicle v : vehicles) {
130             if (v.getVehicleNumber().equals
131                 found = v;
132                 break;
133             }
134         }
135
136         if (found != null) {
137             System.out.println("Vehicle Found");
138             System.out.println("Owner : " +
139                 System.out.println("Claim Amount: ");
140         } else {
141             System.out.println("Vehicle Not Found");
142         }
143
144         sc.close();
145     }
146 }
147
```

Input

```
3
Car
TN10AB1234
Ramesh
```

Output